Muhammad Kashif-Khan

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Personal Profile:

MEng Computer Science and Artificial Intelligence graduate from the University of Leeds with First-Class Honours, specialising in full-stack software architecture, distributed cloud systems, and applied AI. I combine rigorous mathematical foundations with practical engineering experience, ranging from hybrid 3D computer vision pipelines and local LLM orchestration platforms to GPU-accelerated biomedical simulations. I am seeking a graduate role in software engineering, AI, cloud infrastructure, or technical consultancy where I can leverage a versatile skillset across modern software stacks and cloud architectures to deliver immediate technical impact.

Technical Skills:

- **Programming Languages:** Python, C/C++, C#, Java, JavaScript, SQL, Bash
- **AI & Machine Learning:** OpenCV, Scikit-learn, NLP, JAX, Warp, Neural Networks, Llama.cpp, Hermes
- **Web & Cloud Technologies:** Blazor WebAssembly, .NET 8, FastAPI, Azure Functions, Cosmos DB, Redis
- **Scientific Computing & HPC:** CUDA, OpenMP, MPI, NumPy, SciPy, ParaView, CFD, XLB Framework
- **Graphics & Systems:** OpenGL, ROS2, SLAM, PyQt6, Linux/Unix, Docker, Git
- **Specialised Domains:** AI, Robotics, Healthcare, Scientific analysis and research, Finance

Education:

Sep 2022 – Jul 2026 MEng/BSc Computer Science with Artificial Intelligence University of Leeds

Year 2 modules:	Result:	Year 3 modules:	Result:	Year 4 Modules:	Result:
Data Structures and Algorithms	84%	Robotics	86%	Deep Learning	88%
Software Engineering Project	81%	Parallel Computation	76%	Data Science	85%
Software Engineering Principles	78%	Computer Graphics	72%	Advanced S. Engineering	84%
Artificial Intelligence	75%	Machine Learning*	66%	Research Project	84%
Numerical Computation	74%	Distributed Systems*	63%	Cloud Computing	80%

**** Year 3 results affected by extended illness during semester; performance recovered in subsequent modules and projects***

Year 4 - Master's Thesis: Modular 3D Computer Vision Reconstruction Pipeline

- Engineered a hybrid 3D/4D computer vision reconstruction pipeline by integrating iterative 2D point tracking frameworks to generate accurate coordinate trajectories alongside dense depth estimation models.
- Conducted rigorous mathematical optimisation and quantitative benchmarking across standard test scenes, achieving centimetre-accurate error metrics and compiling comprehensive academic report documentation.

Year 3 – Distributed Systems Project: Azure Serverless Inventory Management System

- Architected an event-driven serverless ERP inventory system utilising Azure Functions and Cosmos DB, engineering real-time stock tracking, database change feeds, and automated API-based alert workflows.
- Executed comprehensive system performance benchmarking and optimisation, mitigating FaaS cold start latency and verifying high-throughput concurrent invocation across distributed cloud microservices.

Year 2 – Data Mining Project: Medical AI with GPT-4

- Designed an EPSRC-style research framework integrating GPT-4 with clinical data mining, architecting explainable AI pipelines optimised for NHS workflow automation and clinical decision support.
- Applied the CRISP-DM methodology with strict GDPR compliance, delivering comprehensive deployment feasibility analysis, risk assessments, and cost-benefit modelling for enterprise healthcare integration.

Allerton High Sixth Form College (2020 – 2022) | A-Levels: Mathematics (A), Chemistry (A), Physics (A), Further Maths AS (A), EPQ (A*)

Allerton High School (2017 – 2020) | 11 GCSEs at Grades 9-6 (including Mathematics – 9, English Language – 7)

Work Experience:

June 2025 – Aug 2025 Summer Internship at the University of Leeds – Stent deployment algorithm

- Engineered an advanced physics-driven biomedical simulation platform for flow diverter stent deployments in vascular geometries, building a GPU-accelerated 3D visualisation pipeline using PyVista, VTK, and VMTK.

- Developed joint-level deployment mechanics, collision detection, and force simulation systems to model complex wire mechanics, establishing a scalable foundation for medical device R&D.

June 2024 – Aug 2024 Summer Internship at the University of Leeds – Python-based AFM Viewer

- Programmed the Python-based 'NanoLocz' atomic force microscope data analysis platform using PyQt6 and parallel computing, delivering a 50% performance improvement over the legacy MATLAB codebase.
- Led a cross-functional development team through project planning and task delegation, implementing a modular OOP plugin architecture and mentoring a junior intern to meet core scientific stakeholder requirements.

Personal Technical Projects:

Autonomous AI Agent & Local LLM Orchestration Architecture | Personal Infrastructure

- Architected a local LLM orchestration and autonomous coding agent environment utilising Llama.cpp, Hermes, and containerised microservices across a distributed home-lab server infrastructure (Ubuntu, Raspberry Pi).
- Engineered secure mesh networking and node-to-node communication using Tailscale, enabling **encrypted mobile remote access for on-the-go server management** and optimising sustained inference workloads on high-performance hardware (AMD Ryzen 9 9900X, NVIDIA RTX 5090)

Technical Founder & Software Architect | Project Synergy (AI DevOps Workstation & Observability Platform)

- Conceptualised and architected an advanced developer tooling platform focused on coding agent observability, context preservation across autonomous tasks, and automated DevOps workflow orchestration.
- Designed technical architecture briefs and verification frameworks for tracking autonomous agent execution, optimising multi-agent task delegation and end-to-end software development workflows.

Core Executive Leadership Team | Leeds Artificial Intelligence Society

- Appointed to the core executive leadership team to organise and lead campus technical workshops, manage society operations, and drive student engagement in applied artificial intelligence and software engineering.

Additional Experience:

- **Volunteering at Oxfam (Aug 2021 – March 2022**): Independently managed the shop floor, customer service, and stock inventory while marketing and advertising products online through Oxfam's e-commerce platform.
- **Physics Work Experience at The White Rose University Consortium (June 2021):** Conducted foundational research into quantum computing with light and presented an academic brochure communicating complex physics concepts to university researchers.

Interests and Achievements:

- **UK Mathematics Challenge:** Attained Bronze, Silver, Gold, and Pink Kangaroo awards, demonstrating exceptional analytical reasoning and mathematical problem-solving ability.
- **V100 Volunteering Award:** Recognised for dedicating over 100 hours to community service, operational support, and retail leadership through voluntary work with Oxfam.
- **STEM Tutoring & Mentoring:** Provided structured academic tutoring in mathematics and computer science, developing strong technical communication, pedagogical, and mentoring skills.